

Maxime Denis

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Master of Engineering in Mechanical Engineering candidate (UC Berkeley) with 4 years of hands-on experience in mechanical design (CAD CATIA V5, SOLIDWORKS), product development, and multiphysics simulation (COMSOL, ABAQUS, STAR-CCM+), with direct involvement in industrial production environments and multi-discipline technical documentation. Open to relocate.

Education

University of California, Berkeley

Master of Engineering in Mechanical Engineering (MEng ME)

Berkeley, California, USA

Aug. 2025 – Present

Arts et Métiers ParisTech

Bachelor of Science in Industrial & Mechanical Engineering

Paris, France

Sept. 2023 – June 2025

Skills

Energy & Thermal Systems: Thermodynamics, Fluid Mechanics, Heat Transfer, Cooling Systems, Hydrogen Transport.

Design & Simulation: CATIA V5, SolidWorks, ABAQUS (FEA), STAR-CCM+ (CFD), COMSOL Multiphysics.

Manufacturing & Prototyping: 3D printing, machining, DFM/DFA, prototype testing, production line integration.

Programming & Data: Python, MATLAB, TypeScript, MySQL, SQLite, experimental data processing.

Professional Experience

R&D Engineer Intern – Offshore Hydrogen Transport | Hynamics, Paris

Jan. 2025 – June 2025

- **[System Architecture & Sizing]** Designed liquid hydrogen storage and transfer systems for offshore-onshore transport chains, performing thermodynamic and pressure-drop calculations in Python and Excel to derive equipment specifications and sizing parameters.
- **[Design Adaptation & Optimization]** Iteratively refined system models against technical and operational constraints (boil-off rates, energy consumption, safety margins), producing ranked design concepts and client-ready recommendations under senior engineer supervision.
- **[Technical Deliverables]** Produced engineering notes, sizing calculations, and technical drawings documenting selected architectures, design assumptions, and performance trade-offs for internal and client stakeholders.

Systems Engineer (Part-time) | ORA, Paris

Mar. 2025 – May 2025

- **[Multi-Discipline Drawing Analysis]** Cross-referenced fire device location plans, electrical distribution layouts, and system flow network diagrams to consolidate 500+ assets into a structured engineering database.
- **[System Architecture Mapping]** Reconstructed the functional architecture of fire protection installations from as-built documentation, identifying equipment interdependencies and non-compliant configurations.
- **[Documentation Standards]** Established naming conventions and traceability rules aligned with as-built drawing standards to support future digital twin integration.

Methods / Production Engineering Intern | AsteelFlash, Bedford, UK

June 2024 – July 2024

- **[Component Integration & Layout]** Optimized spatial placement of electronic assembly components within a new production line, resolving integration constraints between stations to ensure smooth material flow and assembly sequence.
- **[Industrial Process Knowledge]** Gained hands-on understanding of production line architecture, takt-time logic, and manufacturing constraints – directly applicable to design-to-manufacture decisions in a design office environment.
- **[Process Documentation]** Authored work instructions and setup checklists from technical specifications to secure line ramp-up and reduce non-conformities during production start.

Selected Projects

R&D Mechanical Design & Simulation Engineer - CO₂-Free Hydrogen Reactor - Capstone Project |

UC Berkeley & Lawrence Berkeley National Lab

Aug. 2025 – Present

- **[Reactor Mechanical Design]** Engineered cavity geometry, waveguide interfaces, and gas injection hardware via parametric CAD modeling to sustain stable microwave plasma for methane-to-hydrogen cracking.
- **[Theoretical & Numerical Validation]** Developed Python-based analytical models to pre-validate design assumptions and derive input parameters — thermal boundary conditions, flow rates, EM field estimates — prior to COMSOL multiphysics simulations.
- **[Thermal & Flow Modeling]** Developed 3D COMSOL multiphysics models coupling EM wave propagation, conjugate heat transfer, and compressible flow to resolve temperature fields and optimize Hydrogen production.

CAD / CFD / FEA Projects | UC Berkeley & Arts et Métiers ParisTech

2022 – Present

- **[Scope]** Heart pump design & CFD optimization (STAR-CCM+, +10% efficiency), aircraft brake system structural validation (ABAQUS), water-filled double-glazing thermal system (<5% model variance), shoulder prosthesis design & FEA.
- Full project details, FEA/CFD results, and prototypes: https://maximedns5.github.io/Portfolio_MD